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PERFORMANCE SPECIFICATION

GENERATOR, CHLORINE – PORTABLE ELECTROLYTIC (PECG)

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CONTENTS

PARAGRAPH PAGE
1. SCOPE 1
1.1 Scope 1
2. APPLICABLE DOCUMENTS 1
2.1 General 1
2.2 Government documents 1
2.2.1 Specifications, standards, and handbooks 1
2.2.2 Other Government documents, drawings, and publications 2
2.3 Non-Government publications 2
2.4 Order of precedence 3
3. REQUIREMENTS 3
3.1 First article 3
3.2 Design conditions 3
3.2.1 Rated condition 4
3.2.2 General operational conditions 4
3.2.3 Environmental conditions 4
3.3 Reliability and maintainability 4
3.4 Design life 4
3.5 Space and weight 4
3.6 Selection of equipment 4
3.7 Electrolytic chlorine generator cells 5
3.7.1 Construction of the generating cells 5
3.8 Shipboard interface 5
3.8.1 PECG hose 5
3.8.2 PECG hose couplings 5
3.8.3 Chlorinator cell electrical isolation and grounding 5
3.8.4 PECG mounting 5
3.8.5 Input power 5
3.8.6 Portable chlorinator unit 5
3.8.7 Auxiliary pump unit 5
3.8.7.1 Performance characteristics 5
3.8.7.2 Motor 5
3.8.8 PECG cabling 6
3.8.9 PECG grounding 6
3.9 Materials 6
3.9.1 Electrolytically dissimilar metals 6
3.9.2 Mercury 6
3.9.3 Painting 6
3.9.4 Noxious fumes and harmful effects 6
3.9.5 Recycled, recovered, or environmentally preferable materials 6
3.10 Threaded fasteners 6
3.11 Shock 6
3.12 PECG electrical requirements 6
3.12.1 Performance ratings 6
3.12.2 AC power 6
3.12.3 Power supply 7

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Check the source to verify that this is the current version before use.

PARAGRAPH PAGE

3.12.4 Main power ON-OFF 7
3.12.5 Enclosures 7
3.12.6 Cooling method 7
3.12.7 Safety ground 7
3.12.8 Insulation resistance 7
3.12.9 AC voltage withstand 7
3.12.10 Electromagnetic interference (EMI) emissions 7
3.12.11 Equipment capacitance and EMI filters 7
3.12.12 Safety devices 7
3.12.12.1 Over-voltage/over-current protection 7
3.12.12.2 Fuses 7
3.12.12.3 Low flow protection 7
3.12.13 Connectors 7
3.12.14 Internal wiring 8
3.12.15 Indicator lights 8
3.12.16 Voltage spike 8
3.13 Safety 8
3.14 Human factors 8
3.15 Interchangeability 8
3.16 Identification marking and label plates 8
3.17 Repair parts and stock spare parts 8
3.18 Drawings 9
4. VERIFICATION 9
4.1 Classification of inspections 9
4.2 First article inspection 9
4.3 Conformance inspection 9
4.4 Visual and dimensional examination 9
4.5 Methods of inspection 10
4.5.1 Insulation resistance 10
4.5.2 General operation and temperature rise 10
4.5.3 Voltage spike 11
4.5.4 Electromagnetic interference 11
4.5.5 Dielectric tests 11
4.5.6 Ground continuity 11
4.5.7 Shock test 11
4.5.8 Performance test 11
4.5.9 Non-destructive testing 11
4.5.10 Hydrostatic testing 11
5. PACKAGING 12
5.1 Packaging 12
6. NOTES 12
6.1 Intended use 12
6.2 Acquisition requirements 12
6.3 Technical manuals 12
6.4 Definitions 12
6.4.1 Auxiliary pump unit 12

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PARAGRAPH PAGE

6.4.2 Portable chlorinator unit 12
6.4.2.1 Chlorinator cells 12
6.4.2.2 Controller 12
6.4.2.3 Power supply assembly 12
6.4.3 Portable electrolytic chlorine generator (PECG) 12
6.5 Subject term (key word) listing 12

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1. SCOPE

1.1 Scope. This specification covers portable electrolytic chlorine generator (PECG) system components intended to provide protection against marine biological fouling (biofouling) in seawater piping systems on Navy Surface Ships.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28	-	Screw-Thread Standards for Federal Services

COMMERCIAL ITEM DESCRIPTIONS

A-A-59004	-	Anti-Galling Compound, Thread Lubricating, Seizing Resistant, and Calcium Hydroxide Containing
A-A-59326	-	Coupling Halves, Quick-Disconnect, Cam-Locking Type

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-E-917	-	Electric Power Equipment, Basic Requirements
MIL-DTL-24640	-	Cables, Light-Weight, Electric, for Shipboard Use, General Specification for
MIL-DTL-24643	-	Cables and Cords, Electric, Low Smoke, for Shipboard Use, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-1310	-	Standard Practice for Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety
MIL-STD-1399-300	-	Interface Standard for Shipboard Systems, Section 300, Electric Power, Alternating Current (Metric)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-237	-	Electromagnetic Environmental Effects and Spectrum Supportability Guidance for the Acquisition Process

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MIL-HDBK-454	-	General Guidelines for Electronic Equipment
MIL-HDBK-2036	-	Preparation of Electronic Equipment Specifications

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVSEA PUBLICATION

NAVSEAINST 5100.3	-	Mercury, Mercury Compounds, and Components Containing Mercury or Mercury Compounds; Control of

(Copies of this document are available from IHS Inc., 15 Inverness Way East, Englewood, CO 80112 or online at www.ihs.com.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN BUREAU OF SHIPPING (ABS)

ABS Guide for Building and Classing Naval Vessels, Part 4: Control, Automation, and Navigation Systems

(Copies of this document are available from the American Bureau of Shipping, Government Operations Office, 1421 Prince Street, Suite 100, Alexandria, VA 22314 or online at www.eagle.org.)

ASME INTERNATIONAL

ASME Y14.24	-	Types and Applications of Engineering Drawings (DoD adopted)
ASME Y14.34M	-	Associated Lists (DoD adopted)
ASME Y14.35M	-	Revision of Engineering Drawings and Associated Documented (DoD adopted)
ASME Y14.100	-	Engineering Drawing Practices (DoD adopted)

(Copies of these documents are available from ASME International, 22 Law Drive, PO Box 2900, Fairfield, NJ 07007-2900 or online at www.asme.org.)

ASTM INTERNATIONAL

ASTM B148	-	Standard Specification for Aluminum-Bronze Sand Castings (DoD adopted)
ASTM F1166	-	Standard Practice for Human Engineering Design for Marine Systems, Equipment and Facilities (DoD adopted)

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., PO Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

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INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

CISPR 16-1-1	-	Measuring apparatus
CISPR 16-1-2	-	Ancillary equipment – conducted disturbances
CISPR 16-1-3	-	Ancillary equipment – disturbance power
CISPR 16-1-4	-	Antenna calibration test sites for 30 to 1000 MHz
CISPR 16-1-5	-	Conducted disturbance measurements
CISPR 16-2-1	-	Conducted disturbance measurements
CISPR 16-2-2	-	Measurement of disturbance power
CISPR 16-2-3	-	Radiated disturbance measurements
CISPR 16-2-4	-	Immunity measurements

(Copies of these documents are available from the IEC Central Office, 3 rue de Varembé, PO Box 131, CH-1211 Geneva 20, Switzerland or online at www.iec.ch.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)

IEEE 45	-	IEEE Recommended Practice for Electric Installations on Shipboard (DoD adopted)

(Copies of this document are available from the Institute of Electrical and Electronics Engineers, Inc., 445 Hoes Lane, PO Box 1331, Piscataway, NJ 08855-1331 or online at www.ieee.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA WD 6	-	Wiring Devices - Dimensional Requirements (DoD adopted)
NEMA ICS 1	-	Industrial Control and Systems: General Requirements (DoD adopted)
NEMA 250	-	Enclosures for Electrical Equipment (1000 Volts Maximum) (DoD adopted)

(Copies of these documents are available from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209 or online at www.nema.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design conditions. The PECG shall be designed for 24 hours of continuous use per day maximum, under naval service conditions for extended periods of time. Reliable in service and accessibility of all parts for inspection, maintenance and repair shall be paramount. IEEE 45 and NEMA ICS 1 shall be used as design guides for the PECG.

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3.2.1 Rated condition. The PECG shall be designed to electrolytically produce equivalent chlorine at a rate of 0.26 lb/hr under the following seawater conditions:

Temperature (°F)	68
Total dissolved solids (TDS) (ppt)	35
Ph	7–8.5
Flow rate (gpm)	7–11 (nom.)
Operating pressure (psig)	110

3.2.2 General operational conditions. The PECG shall operate within the following environmental parameters with some reduced capability from the rated condition point specified in 3.2.1.

Seawater temperature (°F)*	28–105
Total dissolved solids (ppt)	20–38

3.2.3 Environmental conditions. The PECG shall operate satisfactorily within the following shipboard environmental parameters:

Ambient temperature (°F)*	40–122
Ambient relative humidity (%)	0–90

*Equipment may employ a high-temperature protection circuit that can turn the equipment off with an engine room ambient temperature greater than 104 °F; however, the equipment shall be capable of being subjected to a continuous ambient temperature of 122 °F without any degradation.

3.3 Reliability and maintainability. The achievement of optimum reliability shall be the paramount controlling principle in the design, fabrication, and assembly of the PECG. Parts subjected to wear or potential corrosion shall be fabricated from materials so designed. Materials and designs which may not be specifically required herein but whose use could result in improved reliability or additional energy savings shall be considered and offered for potential use in fabricating the PECG.

3.4 Design life. The PECG system components shall be designed for a 2-year design life, with the exception of the generating cells (see 3.7.1) which shall be designed with a minimum electrode life of 4 years.

3.5 Space and weight. Space and weight shall be minimized to the extent possible while maintaining reliability and structural integrity of the PECG equipment. Space and weight limits for the portable chlorinator unit and auxiliary pump shall be as follows:

Portable chlorinator unit:

Maximum dimensions (L x W x H) (in.)	32 x 9 x 10
Maximum weight (lb)	45

Auxiliary pump unit:

Maximum dimensions	1.5 cubic feet
Maximum weight (lb)	40

Deviations from the above dimensions and weights require NAVSEA approval.

3.6 Selection of equipment. The PECG equipment shall be selected to the extent possible from “commercial-off-the-shelf” (COTS) components that have been ruggedized for military applications using guidance from MIL-HDBK-454 and MIL-HDBK-2036.

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3.7 Electrolytic chlorine generator cells. The generating cell(s) provide for a once-through chlorine generating capability. The generating cell(s) construction must be of the Concentric Tube Electrode Type to eliminate the need for acid cleaning of the cathode surfaces. Plate type electrodes shall not be allowed. The cell(s) shall contain anodic and cathodic surfaces such that the electrolytic processes occur within the annular surface region between the anode and cathode.

3.7.1 Construction of the generating cells. The structural components of the generating cell(s) shall be PVC. The electrodes shall be commercially pure, Grade 2 (CP GR 2) titanium and the anodes shall have a commercially pure platinum or other acceptable coating with a minimum electrode life of 4 years.

3.8 Shipboard interface. The PECG shall be compatible with the following shipboard interface points as shown in Figure 1 and as follows.

3.8.1 PECG hose. Hose, used as part of the PECG or to interface to shipboard connections, shall be ¾-inch NPS. Hoses selected shall be capable of withstanding internal pressures and seawater conditions listed in 3.2 over the life of the system. Hose shall be capable of withstanding continuous exposure to chlorinated seawater (up to 50 ppm total chlorine).

3.8.2 PECG hose couplings. Hose couplings shall conform to A-A-59326 and shall be capable of withstanding internal pressures and seawater conditions listed in 3.2 over the life of the system. Couplings shall be capable of withstanding continuous exposure to chlorinated seawater (up to 50 ppm total chlorine).

3.8.3 Chlorinator cell electrical isolation and grounding. Chlorinator cell shall be isolated from the shipboard piping system. Personnel contact with portions of the chlorinator cells capable of exhibiting a voltage potential to ground shall be prevented through the use of electrically isolated protective covers or an enclosure.

3.8.4 PECG mounting. Due to the portable nature of the PECG components shipboard, mounting of both the chlorinator unit and the auxiliary pump unit to mounting brackets shall be accomplished by the use of strapping and/or tie downs appropriate for use shipboard. Any mounting system utilized shall be capable of surviving the Grade B, Type A shock test (see 3.11 and 4.5.7).

3.8.5 Input power. The portable chlorinator unit and the auxiliary pump unit shall be compatible with the AC input power requirements (see 3.12.2).

3.8.6 Portable chlorinator unit. The portable chlorinator unit shall be installed downstream of the auxiliary pump unit. The portable chlorinator unit shall have dimensions to adequately fit into the dimensions outlined in 3.5, using the mounting arrangements of 3.8.4. The use of flexible hose and connections are permitted provided they meet the pressure requirements of 3.2.1. The portable chlorinator unit interfaces to the shipboard piping at the isolation valve provided by the installation support activity which connects to the seawater system injection location (see Figure 1).

3.8.7 Auxiliary pump unit. The auxiliary pump unit shall be provided upstream of the generator. The pump shall be close-coupled, closed impeller, centrifugal pump. The pump shall be made of the materials suitable for seawater service onboard surface ships. The wetted end shall be nickel aluminum bronze in accordance with ASTM B148, C95800. The use of flexible hose and connections are permitted provided they meet the pressure requirements of 3.2.1. The auxiliary pump unit shall have dimensions to adequately fit into the dimensions outlined in 3.5, using the mounting arrangements of 3.8.4. The auxiliary pump unit interfaces downstream of the Y-strainer (supply system strainer) (see Figure 1).

3.8.7.1 Performance characteristics. The minimum rated total head of the pump shall be 35 feet at the rated capacity of 9 gpm. The fluid to be pumped will be seawater at the conditions specified in 3.2.1 and 3.2.2.

3.8.7.2 Motor. The pump shall be driven with an AC continuous duty motor, which shall be UL listed. The pump's motor shall operate on MIL-STD-1399-300, Type I, 115 VAC, 1-phase, 60-Hz power source. The rated horsepower of the pump motor shall be equivalent to at least the maximum brake horsepower of the driven pump under all conditions from shutoff to free delivery. Head capacity and brake horsepower curve shall demonstrate that the motor is not overloading at any point. The pump motor shall be rated for 50 °C ambient temperature.

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3.8.8 PECG cabling. The PECG system design load current ranges are noted below. Cables to service these loads may be selected from MIL-DTL-24640 or MIL-DTL-24643 or may be COTS cables using the cited specifications as a guide for conformance. The method of electrical cable termination shall be compatible with the cable types selected and MIL-STD-1399-300, 115 V, single-phase Type I power for portable equipment.

a. PECG power input: 15 amps, 115 VAC, single-phase, 60 Hz.

b. Power supply output: [one (+), one (-)] 0-50 amps manually adjustable, 0-15 VDC automatically varied by the power supply to maintain constant current setting – output voltage limit to not exceed 15 VDC.

3.8.9 PECG grounding. The PECG shall be a safety grounded in accordance with MIL-STD-1310 through the power cable isolated safety lead and power plug safety pin.

3.9 Materials.

3.9.1 Electrolytically dissimilar metals. Direct contact of electrolytically dissimilar metals shall be avoided where destructive electrolysis would result.

3.9.2 Mercury. The intended use makes it mandatory that material furnished be free of any mercury contamination. Control of mercury and mercury containing compounds shall be in accordance with NAVSEAINST 5100.3. During the manufacturing processes, tests and inspections, the product offered for acceptance shall not have come in direct contact with any mercury containing device employing a single boundary of containment. The manufacturer shall certify the product shipped is free from mercury contamination.

3.9.3 Painting. Individual component painting and finishing shall be suitable to withstand environmental and operating conditions identified in this specification and related component specifications. Painting shall be in accordance with MIL-E-917 or commercial specifications. Powder coating may be used in accordance with commercial specifications. Paint shall not be applied to corrosion resistant materials including corrosion resistant steel, nickel-copper or copper-nickel alloys, nickel-chromium-iron alloy, nickel-chromium-molybdenum-columbium alloy, titanium, brass, bronze, or copper.

3.9.4 Noxious fumes and harmful effects. Fittings, paint, insulation, adhesives, or other items containing material or components that would give off noxious fumes at any temperature below 200 °F shall not be installed or applied. For paint and adhesives, this requirement applies after drying or curing is complete.

3.9.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.10 Threaded fasteners. Threaded parts, such as bolts, studs and nuts, shall be unified thread series and conform to FED-STD-H28. Stainless steel threaded fasteners used to fasten to stainless steel components or female thread assemblies shall utilize anti-galling compound in accordance with commercial item description A-A-59004.

3.11 Shock. The PECG system components shall be shock qualified for high impact, lightweight, Grade B, Type A shock in accordance with MIL-S-901. The shock test shall be equipment Class I. The PECG shall be mounted to simulate actual shipboard arrangement.

3.12 PECG electrical requirements. PECG electrical equipment shall be selected to the extent possible from “commercial-off-the-shelf” (COTS) components that have been ruggedized for military applications using guidance from MIL-HDBK-454 and MIL-HDBK-2036. The PECG design shall follow the basic schematic shown in Figure 2. All electrical equipment shall meet the following requirements.

3.12.1 Performance ratings. Power supply output DC volts and DC amperes shall be such that chlorine production meets the requirements specified in 3.2.1.

3.12.2 AC power. The PECG’s input power requirements shall be compatible with Type I, ungrounded, 60-Hz, shipboard power characteristics as defined in MIL-STD-1399-300. Unless otherwise specified (see 6.2), the power supply shall operate from a 115-VAC, 60-Hz, 1-phase input power source and shall not require more than 15 Amps current, power connection made with a 3-prong plug that meets NEMA 5-15P or equivalent in accordance with ANSI/NEMA WD 6.

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3.12.3 Power supply. The power supply shall be the constant current type designed to maintain amperage to the chlorinator cells at a preset level during the seawater conditions specified in 3.2.1. The output current shall be adjustable to maintain the rated chlorine generation level specified in 3.2.1 over the widest range of operational conditions specified in 3.2.1 and 3.2.2 as practical. An adjustment control feature shall be designed to be capable of manually adjusting for changes in seawater conditions, and it shall be located on the front of the controller/power supply assembly at an easily accessible location. Some degradation in performance will be acceptable at the extreme ends of the operating range (see 3.2.2 for additional requirements). The DC output current shall be automatically regulated within ±5 percent of the selected current output throughout the current output selection range for all combinations of input power as defined in 3.12.2, ambient temperature as defined in 3.2.3, and changes in load within the aforementioned change adjustment range.

3.12.4 Main power ON-OFF. The PECG shall be provided with a main power ON-OFF switch located on the portable chlorinator unit enclosure and a separate pump ON-OFF switch on the auxiliary pump unit enclosure. The switches shall open all conductors except the safety ground. The switches shall be rated for at least 15 amps. The power ON-OFF switches shall be clearly labeled.

3.12.5 Enclosures. The PECG enclosures shall be drip-proof protected to 45 degrees using MIL-STD-108, as a guide. All electrical enclosures shall be NEMA 250 or equivalent.

3.12.6 Cooling method. The cooling method should be natural convection nonventilated or natural convection ventilated. Forced air ventilation is also acceptable. External enclosure surface temperature shall not exceed the requirements of MIL-HDBK-454, Guideline 1.

3.12.7 Safety ground. All accessible surfaces of the portable chlorinator unit or the auxiliary pump unit shall be at ground potential (also see 3.8.9). In addition, hinged doors (if used) with electrical components mounted on the door shall use flexible ground straps to provide the ground connection to the assembly enclosure.

3.12.8 Insulation resistance. Insulation resistance of the equipment shall not be less than 10 megohms at 25 °C, measured at 500 VDC. When exposed to a 500 V megger, all components should be suitably protected. Measurement shall be made between all mutually insulated points and between insulated points and ground.

3.12.9 AC voltage withstand. The PECG shall be designed and constructed to withstand the application of an 1230 VAC (rms) potential between electrically conductive parts and ground for one minute (twice rated +1000 V).

3.12.10 Electromagnetic interference (EMI) emissions. The PECG shall be designed using MIL-HDBK-237 as a guide to minimize conducted and radiated emissions within the limits of the ABS tests specified in 4.5.4.

3.12.11 Equipment capacitance and EMI filters. The use of line-to-ground filters for EMI control shall be minimized. If such filters are required, the line-to-ground capacitance for each line shall not exceed 0.1 microfarads. Line-to-line filters are the preferred method of filtering. EMI filters shall be located on the load side of the power ON-OFF switch when practical to minimize capacitive loading of the electrical system, to prevent failure of EMI filters when equipment is not in use, and to prevent potential resonant conditions.

3.12.12 Safety devices. The following safety devices shall be provided as a minimum for the PECG.

3.12.12.1 Over-voltage/over-current protection. The power supply shall be provided with over-voltage and over-current protection.

3.12.12.2 Fuses. Both the portable chlorinator unit and the auxiliary pump unit shall be provided with power fuses (see Figure 2). All fuses shall be panel-mounted and accessible from the front of the PECG enclosure.

3.12.12.3 Low flow protection. The PECG shall provide a low-flow cutout circuit that de-energizes the electrolytic cell DC power supply at chlorinator flow rates less than 7 gpm. Upon loss of power due to low flow conditions, the power supply shall remain de-energized until flow has been re-established. At flow rates greater than 7 gpm and less than 11 gpm, the DC power supply shall be energized.

3.12.13 Connectors. Male plug connectors shall be wired to be de-energized when unplugged. Metal shell of multi-pin connectors shall be grounded.

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3.12.14 Internal wiring. The minimum temperature rating for all wiring shall be 90 °C. The minimum wire size shall be AWG 14, except that AWG 16 may be used for connections to local switches and that, at a minimum, AWG 22 may be used for all connections to the controller and selected solder connected devices as appropriate for device termination.

3.12.15 Indicator lights. Color designated for indicator lights shall be as follows:

*Red	Danger or emergency requiring immediate attention
Green	Normal condition
White	Power on or power available
Blue	Closed or shut
Yellow	Special
Amber	Abnormal but not immediately dangerous

*Indicator lights shall be wired so that the light becomes energized upon abnormal conditions.

3.12.16 Voltage spike. The PECG shall be designed and constructed to withstand the application of voltage spike test in accordance with MIL-STD-1399-300 with a waveform specified in Figure 2 of that standard.

3.13 Safety. The PECG shall present no uncontrolled hazards to operating or maintenance personnel.

3.14 Human factors. All man-to-machine interfaces (i.e., controls, displays, alarms, etc.) shall be suitable for user personnel with applicable fifth and ninety-fifth percentile body dimensions as defined in ASTM F1166.

3.15 Interchangeability. All units shall be interchangeable from the same manufacturer. All component parts, including stock spare parts shall be interchangeable without adapters, selective assembly or further machining. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of ASME Y14.24, ASME Y14.34M, ASME Y14.35M, and ASME Y14.100.

3.16 Identification marking and label plates. PECG shall be labeled. Labels may be supplied to any applicable commercial specification as long as the label is permanently applied to the components, is legible, and prominently displayed. PECG label plates shall include the following:

a. Component title
b. Manufacturer's name
c. Serial number
d. Date (month and year) of manufacture
e. Drawing number
f. Technical manual number, as applicable
g. National stock number, as applicable

3.17 Repair parts and stock spare parts. Repair parts and stock spare parts shall be as specified (see 6.2).

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3.18 Drawings. Drawings shall be prepared when specified (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one complete portable electrolytic chlorine generator when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.8 in the order shown in Table I.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.1, 4.5.2, 4.5.6, 4.5.9, and 4.5.10 in the order listed.

4.4 Visual and dimensional examination. Each PECG shall be visually and dimensionally inspected to verify compliance with drawing and specification requirements not involving tests.

a. The overall dimensions and weights do not exceed the maximum values specified in 3.5.

b. Components are mounted as shown on the approved drawings.

c. Components are marked with identifying symbols and these symbols agree with those on the approved drawings.

d. Terminals of all components and terminal boards are marked with identifying letters or numbers or both, and this identification agrees with those used on the approved drawings.

e. Wire used for interconnecting components is of the type shown on the approved drawings.

f. Cable entrance provisions are as shown on the approved drawings.

g. Both ends of all wires are marked with designations as shown on the approved drawings.

h. Identification and instruction plates are furnished as shown on the approved drawings.

i. The generating cells utilized are in accordance with the type specified in 3.7 and 3.7.1.

j. The hose selected is in accordance with the details of 3.8.1.

k. The hose coupling selected are in accordance with 3.8.2.

l. The chlorinator cells are isolated from the shipboard piping system in accordance with 3.8.3.

m. The PECG mounting system is in accordance with 3.8.4.

n. The input power selected for the PECG is in accordance with 3.8.5.

o. The pump selected for the PECG is in accordance with 3.8.7.1.

p. The motor driving the pump for the PECG is in accordance with 3.8.7.2.

q. The PECG cabling has been selected in accordance with 3.8.8.

r. The materials selected for the PECG are in accordance with 3.9.

s. The threaded fasteners selected for the PECG are in accordance with 3.10.

t. The electrical equipment selected for the PECG should be COTS and ruggedized in accordance with 3.12.

u. The power supply selected for the PECG has performance ratings in accordance with 3.12.1.

v. The PECG power requirements are in accordance with 3.12.2.

w. The power supply selected for the PECG has all of the features of 3.12.3.

x. The main power ON-OFF features are in accordance with 3.12.4.

y. The enclosures selected for the PECG are in accordance with 3.12.5.

z. The cooling method selected for the PECG is in accordance with 3.12.6.

aa. The use of line-to-ground capacitance is in accordance with 3.12.11.

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bb. The safety devices selected for the PECG are in accordance with 3.12.12.

cc. The internal wiring of the PECG is in accordance with 3.12.14.

dd. The indicator lights selected for the PECG are in accordance with 3.12.15.

ee. The man-to-machine interfaces of the PECG are in accordance with 3.14.

ff. Parts selected for the PECG are in accordance with 3.15.

gg. The labeling of the PECG is in accordance with 3.16.

TABLE I. Test procedures.

Description	Requirement paragraph	Verification paragraph	First article inspection	Conformance inspection
Visual and dimensional examination	3.3, 3.4, 3.5, 3.7, 3.8.1, 3.8.2, 3.8.3, 3.8.4, 3.8.5, 3.8.6, 3.8.7.1, 3.8.7.2, 3.8.8, 3.9, 3.10, 3.12, 3.12.1, 3.12.2, 3.12.3, 3.12.4, 3.12.5, 3.12.6, 3.12.11, 3.12.12, 3.12.14, 3.12.15, 3.14, 3.15, & 3.16	4.4	X	X
Insulation resistance	3.12.8	4.5.1	X	X
General operation and temperature rise	3.2.2 & 3.2.3	4.5.2	X	X
Voltage spike	3.12.16	4.5.3	X	
Electromagnetic interference	3.12.10	4.5.4	X	
Dielectric	3.12.9	4.5.5	X	
Ground continuity	3.8.9, 3.12.7, & 3.12.13	4.5.6	X	X
Shock	3.11	4.5.7	X	
Performance	3.8.6 & 3.8.7	4.5.8	X	
Non-destructive	3.13	4.5.9	X	X
Hydrostatic	3.2.1	4.5.10	X	X

4.5 Methods of inspection. Each PECG chlorinator shall be subjected to the following inspections and tests.

4.5.1 Insulation resistance. Insulation resistance shall be measured to ensure compliance with 3.12.8, in accordance with an approved test procedure. Measurement shall be made between all mutually insulated points and between insulated points and ground. Test voltage shall be applied for at least 60 seconds. Measurements shall be made at any convenient ambient temperature with all circuits cold. Relative humidity and ambient temperature shall be recorded.

4.5.2 General operation and temperature rise. Tests shall be conducted in accordance with an approved test procedure with the equipment completely assembled and mounted in the manner for which it is designed.

a. General operation tests shall be conducted to verify that the required operating features of the PECG function properly.

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b. Temperature rise tests shall be made with the PECG completely assembled and mounted in the manner for which it is designed. If the equipment will produce a higher temperature at other than rated power, it shall be run at the loading which produces the maximum heating. For all components, it shall be determined that the adjacent ambient within the enclosure does not exceed the rated ambient of the individual equipment.

(1) Temperature rise tests shall be performed under conditions equivalent to normal operating conditions at rated voltage, frequency, and load.

(2) The ambient temperature shall not vary by more than 10 °C during test.

4.5.3 Voltage spike. The power supply shall be subjected to a voltage spike test following an approved test procedure. The voltage spike shall be 1000 volts with a waveform as shown in Figure 2 of MIL-STD-1399-300. The power supply shall not be damaged by this test.

4.5.4 Electromagnetic interference. The PECG shall be tested in accordance with the ABS Guide for Building and Classing Naval Vessels, Part 4, Radiated Electric Field Emission test and Conducted Emission test – "For equipment installed in the general power distribution zone". The test procedure will be according to CISPR 16-1-1, 16-1-2, 16-1-3, 16-1-4, 16-1-5, 16-2-1, 16-2-2, 16-2-3, and 16-2-4.

4.5.5 Dielectric tests. Dielectric tests shall be performed after all other electrical tests have been completed. If the insulation resistance for the windings is known to be lower than specified, due to dirt, moisture, or damage to winding, this shall be remedied before the application of the dielectric test voltage.

a. Test voltage. The frequency of the testing voltage shall be not less than 60 Hz and shall approximate a true sine wave. The values of test voltage shall be as specified in 3.12.9, and shall be applied continuously for a period of one minute.

b. Measurement of test voltage. The measurement for the voltage used in dielectric tests shall be made by the voltmeter method whereby the meter derives its voltage from the high-voltage circuit either directly or by means of a voltmeter coil placed in the testing transformer, or through an auxiliary ratio transformer.

c. Points of application. The test voltage shall be successively applied between each electric circuit and all other electrical circuits and grounded metal parts not electrically connected to it. Semiconductor components may be shorted out or disconnected for the dielectric test.

d. Operational test. The PECG shall not be damaged by the performance of this test.

4.5.6 Ground continuity. The purpose of the ground continuity test is to verify that all conductive parts of the PECG that are exposed to user contact are connected to the power line ground (the "green" wire) and the 3-prong power cord. This test shall be performed using a low-current DC source (<1 amp) to determine that there is a low resistance between the ground blade on the power cord and any exposed metal of the PECG.

4.5.7 Shock test. The PECG shall be subjected to a Grade B high-impact shock test in accordance with 3.11. The PECG will be powered up, but not in the operating mode. No external portion of the PECG shall come adrift or otherwise create a missile hazard.

4.5.8 Performance test. Prior to shock testing, the PECG components shall be performance tested to verify the ability to electrolytically produce equivalent chlorine at a rate of 0.26 lb/hr. With the PECG at nominal operating conditions of greater than 7 gpm flow rate, 68±5 °F seawater temperature, and a total dissolved solids concentration of approximately 35 ppt salinity, the unit shall produce at least 0.26 pound of equivalent chlorine per hour for a continuous test period of 2 hours. Failure of this criterion shall be cause for rejection (the PECG is not required to be pressurized to operating pressure for this test).

4.5.9 Non-destructive testing. Non-destructive testing shall be in accordance with basic material specifications listed herein and on the approved drawings.

4.5.10 Hydrostatic testing. Each PECG shall be hydrostatically pressurized with freshwater at ambient temperature to 1½ times the design operating pressure specified in 3.2.1. External leakage, any signs of visual damage, or permanent deformation shall constitute failure.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The PECG equipment covered by this specification is intended for surface ship seawater cooling system use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Quantity required.

c. Type and drawing number (i.e., designator) of manufacturer's drawing required (see 3.18).

d. When first article is required (see 3.1).

e. Design life of replaceable parts if other than 2 years (see 3.4).

f. Power supply if other than 115-VAC, 60-Hz, 1-phase input power source (see 3.12.2).

g. Requirements for repair parts and stock spare parts (see 3.17).

h. When drawing preparation is required (see 3.18).

i. Packaging requirements (see 5.1).

j. Technical manual requirements when specified (see 6.3).

6.3 Technical manuals. Technical manuals should be furnished when specified (see 6.2).

6.4 Definitions. The following terms used throughout this specification refer to the various parts of the chlorinator and its supporting systems.

6.4.1 Auxiliary pump unit. Consists of a pump and associated components. Pump unit operation is controlled by the controller.

6.4.2 Portable chlorinator unit.

6.4.2.1 Chlorinator cells. Consists of chlorine generation electrode cells.

6.4.2.2 Controller. Provides the control signal for the power supply to energize/de-energize during start-up, shutdown during normal and low flow conditions.

6.4.2.3 Power supply assembly. Consists of DC power supply that electrically drives the chlorinator.

6.4.3 Portable electrolytic chlorine generator (PECG). Consists of all equipment of the anti-biofouling system including: portable chlorinator unit (chlorinator cells, electrical controller/power supply assembly) and the auxiliary pump unit.

6.5 Subject term (key word) listing.

Biofouling

Chlorination

Chlorinator

Electrolysis

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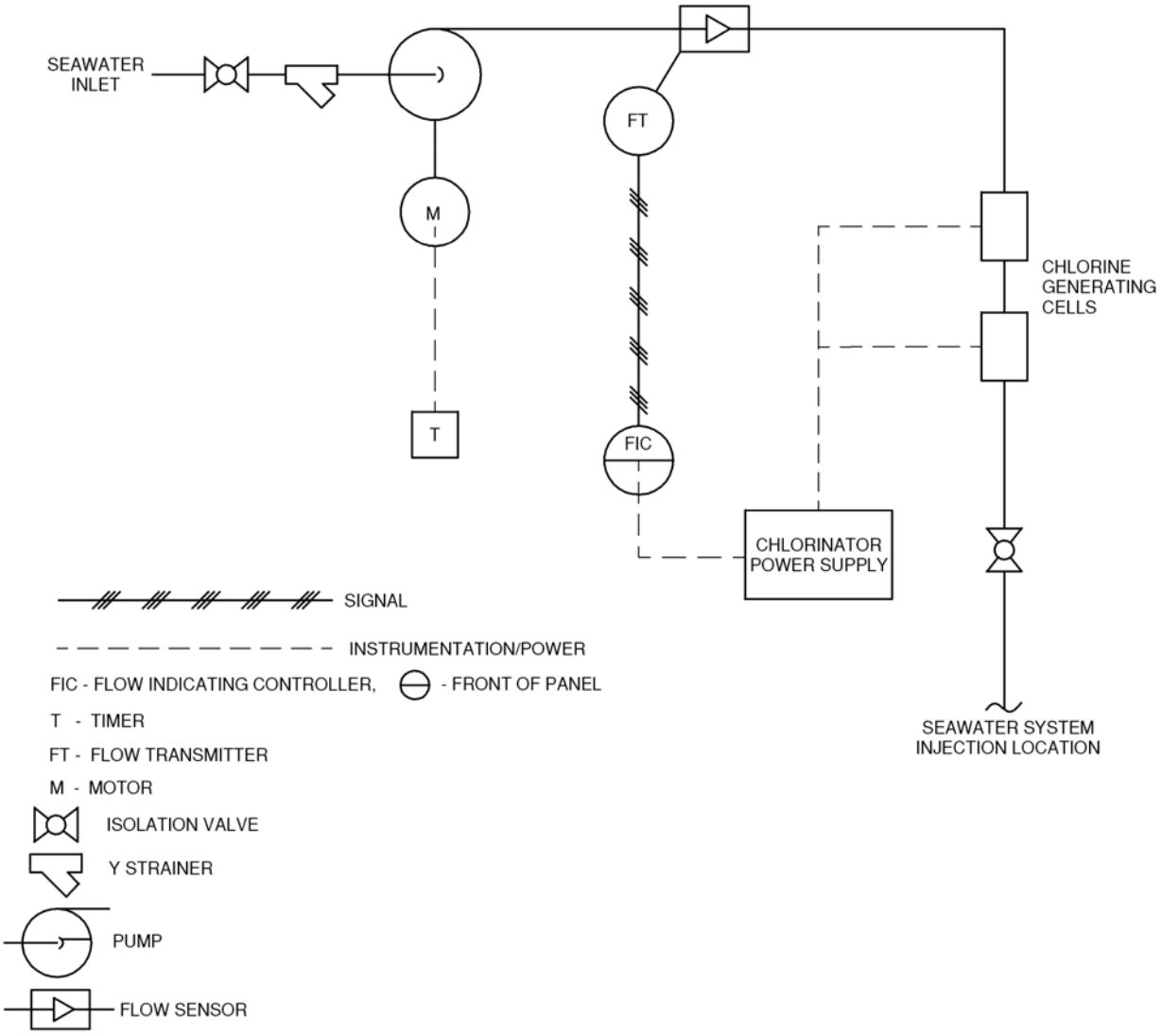


FIGURE 1. Piping and instrumentation drawing.

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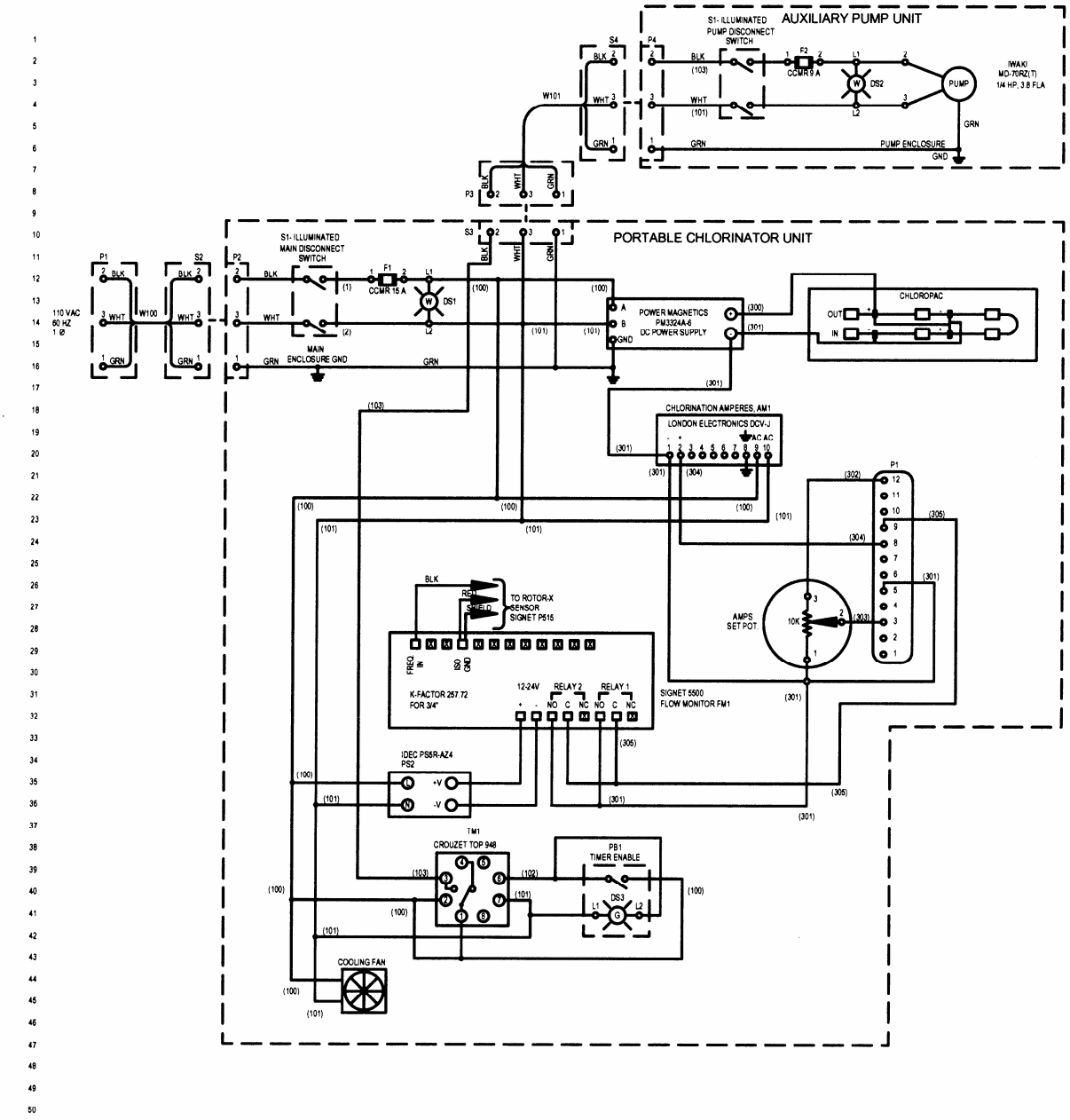


FIGURE 2. PECG electrical system schematic (typical).

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